

Faraz Mazhar

Managing Consultant — Data & AI Engineering
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PROFESSIONAL SUMMARY

Data engineer with 8+ years building scalable data platforms on AWS and Azure. Architected and led development of a modular data framework powering enterprise analytical and ML pipelines. Owns the full lifecycle from data modeling and lakehouse design through API integration, orchestration, and production deployment. Now building agentic platforms — metadata-driven designs with a dynamic orchestrator that composes workflows at runtime.

TECHNICAL SKILLS

AI & Agents	Multi-agent orchestration, dynamic workflow composition, metadata-driven agent frameworks, LLM tool calling
Cloud	AWS (S3, Lambda, Glue, EMR, API Gateway, Step Functions, CloudFormation, DynamoDB, RDS, Athena, CloudWatch, IAM, KMS, SNS, DMS), Azure (Data Factory, Databricks, Functions, EventHub, App Service, SQL DB, Active Directory)
Big Data	Apache Spark, Delta Lake, Databricks, Apache Airflow, Kafka, Hadoop (Hive, HDFS, Yarn), Trino/Presto, Sqoop
Languages	Python (Django, Flask, Pandas, PySpark), SQL, Bash, PowerShell, C/C++
Databases	SQL Server, PostgreSQL (RDS, Redshift), MongoDB, DynamoDB
APIs & IaC	REST, GraphQL (Strawberry-Django), API Gateway, Kinesis, CloudFormation, Git, Linux

EXPERIENCE

Systems Limited Mar 2021 — Present

Managing Consultant

Multi-Agent Orchestration · Python · PySpark · Delta Lake · Databricks · Airflow · Django

- **Agentic Framework:** architected and built a metadata-driven agent framework in which each agent's bespoke logic is isolated to its own module while cross-cutting behavior is driven by metadata — onboarding a new agent is a focused implementation plus configuration, with the bulk of the platform reused across the fleet.
- Owned the full agent lifecycle from boot and registration through invocation, giving the platform end-to-end control over how agents are configured, executed, and observed at runtime.
- Built the framework's flagship capability: a dynamic orchestrator that composes workflows at runtime from the capabilities exposed by registered agents, solving problems for which no pre-defined workflow exists.
- Designed the capability and metadata model that lets agents advertise what they can do — the basis for discovery, composition, and reuse across the agent fleet.
- **Data Framework:** led solution architecture and development of a modular Data Framework (Python, Pandas, PySpark, Delta Lake) for ingestion, curation, and analytics — enabling data pipelines without deployment cycles.
- Owned end-to-end solution architecture across the platform — designed features spanning data ingestion, migration, reconciliation, policy frameworks, alerting, automated reporting, visual analytics, and platform-wide enhancements and optimizations.
- Designed a DSL-driven Canonical Model and Semantic Model for schema-agnostic data curation across arbitrary inputs and outputs.
- Data-modeled the metastore, Delta Lake, and SQL Server datastore; optimized for query performance and multi-tenancy.
- Architected REST and GraphQL APIs for Integration and Semantic services, enabling external systems to submit workloads and consume metadata dynamically.
- Orchestrated multi-stage pipelines with Databricks Workflows, Apache Airflow, and Azure Data Factory; built a dialect-agnostic SQL query builder.

Northbay Solutions Aug 2018 — Mar 2021

Senior Software Engineer

Python · PySpark · AWS Glue · Lambda · Step Functions · CloudFormation · Flask

- Migrated legacy SAS code to PySpark and built reusable packages for AWS Glue ETL jobs; developed a Data Lake framework for large-scale ingestion and processing.
- Built event-driven ETL triggers using Lambda and CloudWatch on S3 events, and API Gateway endpoints for Kinesis Data Streams with DynamoDB tooling.
- Designed dynamically parallel Step Functions State Machines and built an automated cloud migration framework with CloudFormation IaC.
- Developed REST APIs with Flask and a customizable XML parser for document processing.

Mentor Graphics 2017

Intern — Quality Assurance

C/C++ · Python · Bash · Nucleus RTOS

- Performed QA and test automation for Nucleus RTOS; wrote test scripts in Python and Bash.

EDUCATION

BS (Hons.) Computer Science

Punjab University College of Information Technology (PUCIT)
2014 — 2018

CERTIFICATIONS

Microsoft Azure Fundamentals (AZ-900)

2026

AWS Certified Developer — Associate

2018 — 2021